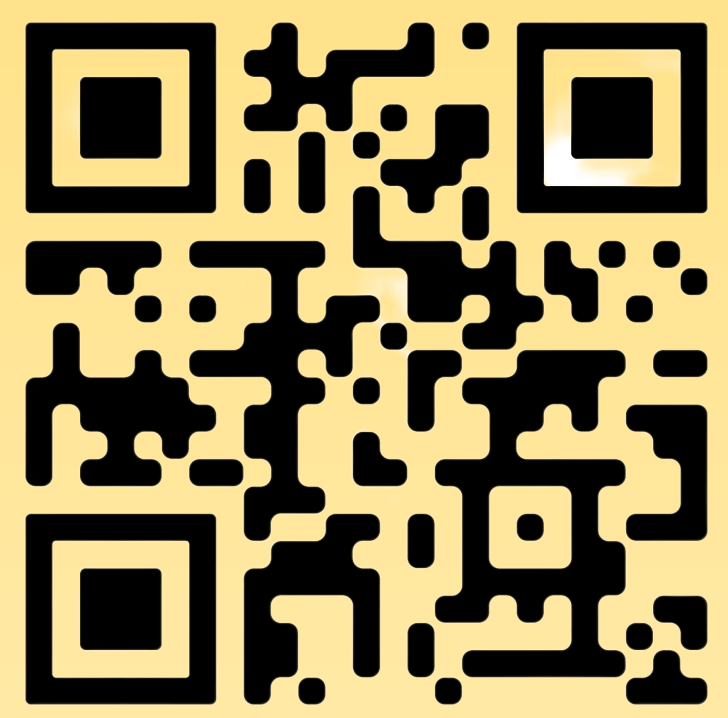


Inceptive Transformers: Enhancing Contextual Representations through Multi-scale Feature Learning Across Domains and Languages

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PAPER

Problem Definition

- Encoder models compress all information from a text sequence into a single [CLS] token representation, which is used almost exclusively in downstream tasks like classification
- Over-reliance on this single token risks creating an information bottleneck as local patterns and fine-grained features are often diluted or averaged out
- The problem manifests in tasks like irony detection, where specific cues (e.g. sarcastic keywords) are just as important as global context
- The issue is even more pronounced in multi-label tasks, which require token-level attention rather than a single sequence-level summary

Stepping towards a Solution

- To remedy this, we need to extract important local patterns from the transformer's token embeddings
- But WHICH patterns are important? A single sarcastic hashtag? A two-word phrase? Or a longer sub-clause?
- Instead of trying to pinpoint certain local cues, we use an inception module to capture features at multiple scales simultaneously—from token-level relationships to phrase-level dependencies
- Convolution works here because it operates on token embeddings that are already rich with global context from the transformer, allowing the inception module to focus exclusively on local patterns that may not have surfaced in the original embeddings

Datasets & Baselines

Task	Dataset	#Texts	#Classes
Irony detection	TweetEval	4,601	2
Emotion recognition	TweetEval	5,052	4
Bangla emotion recognition	Bangla-Banglish	80,098	6
Disease identification	OHSUMED	13,929	23
Anti-covid vaccine concerns classification	CAVES	9,921	11

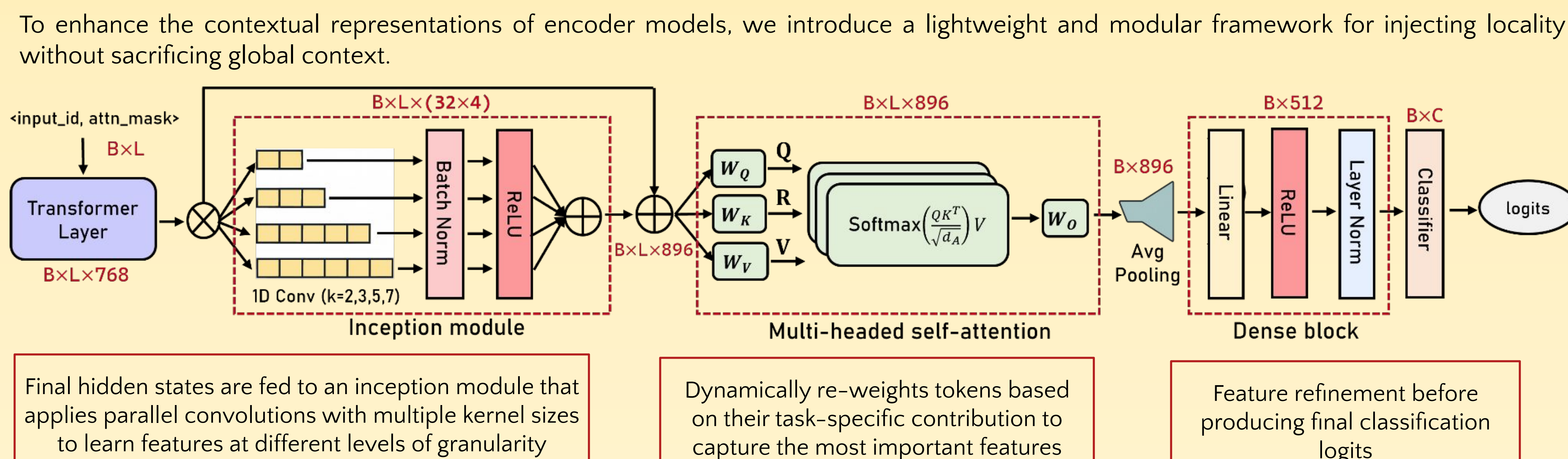
General models:

- RoBERTa
- DeBERTa
- ModernBERT
- XLNet

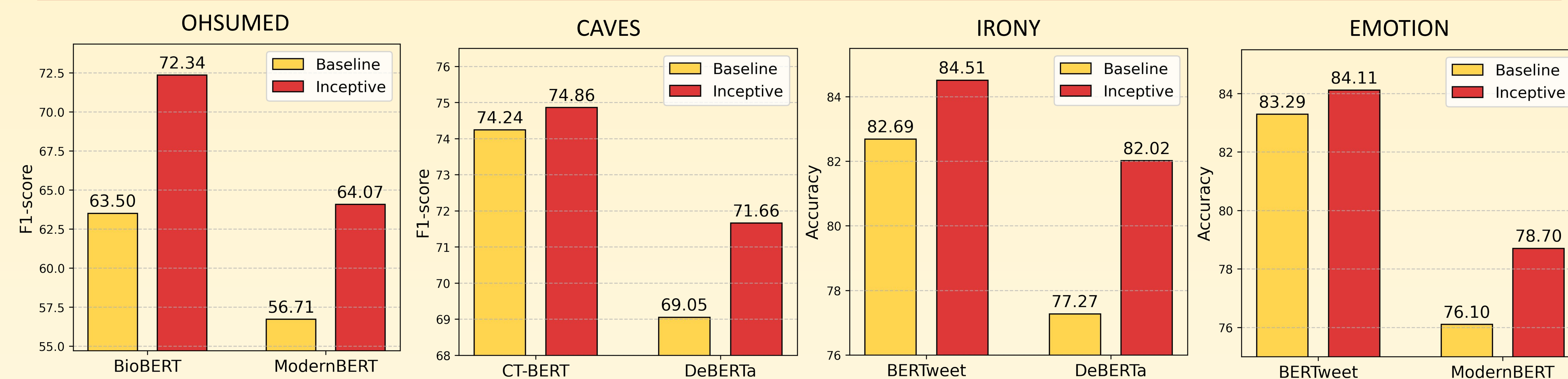
Domain-specific models:

- BERTweet
- BioBERT
- CT-BERT
- BanglaBERT

Proposed Architecture



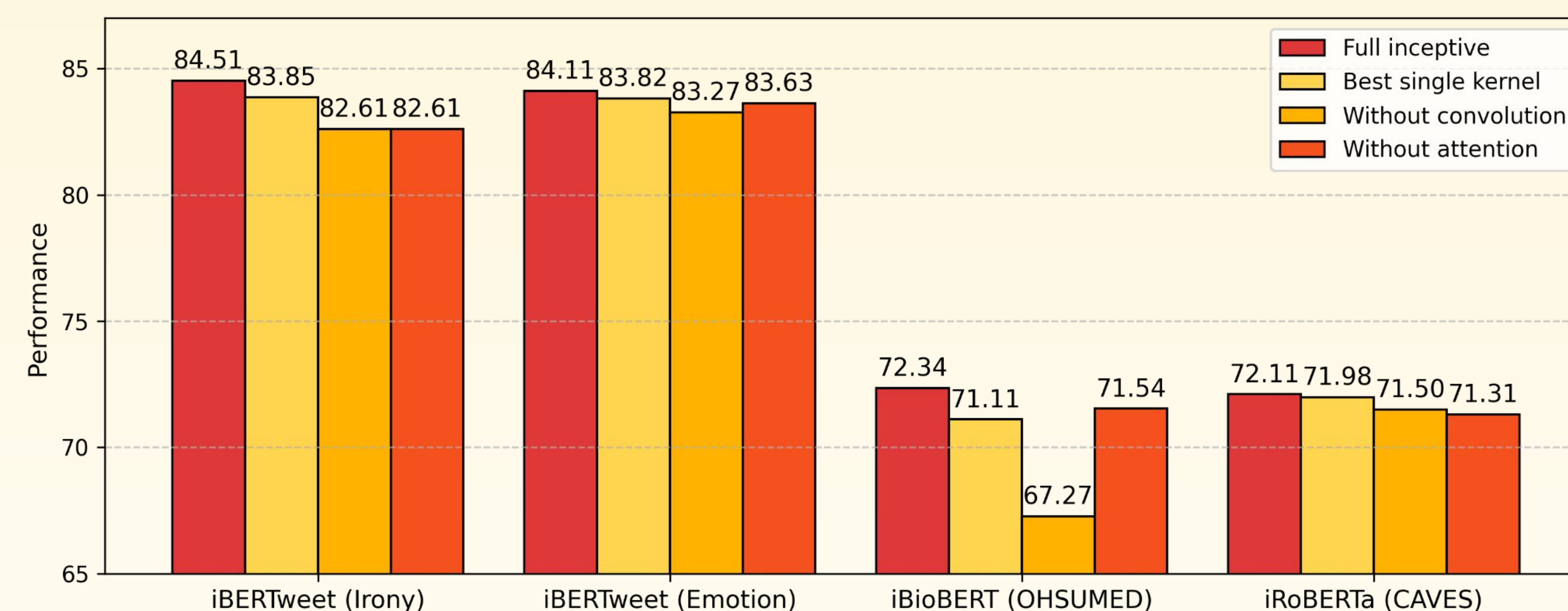
Comparative Reslts, Ablation, and Analysis



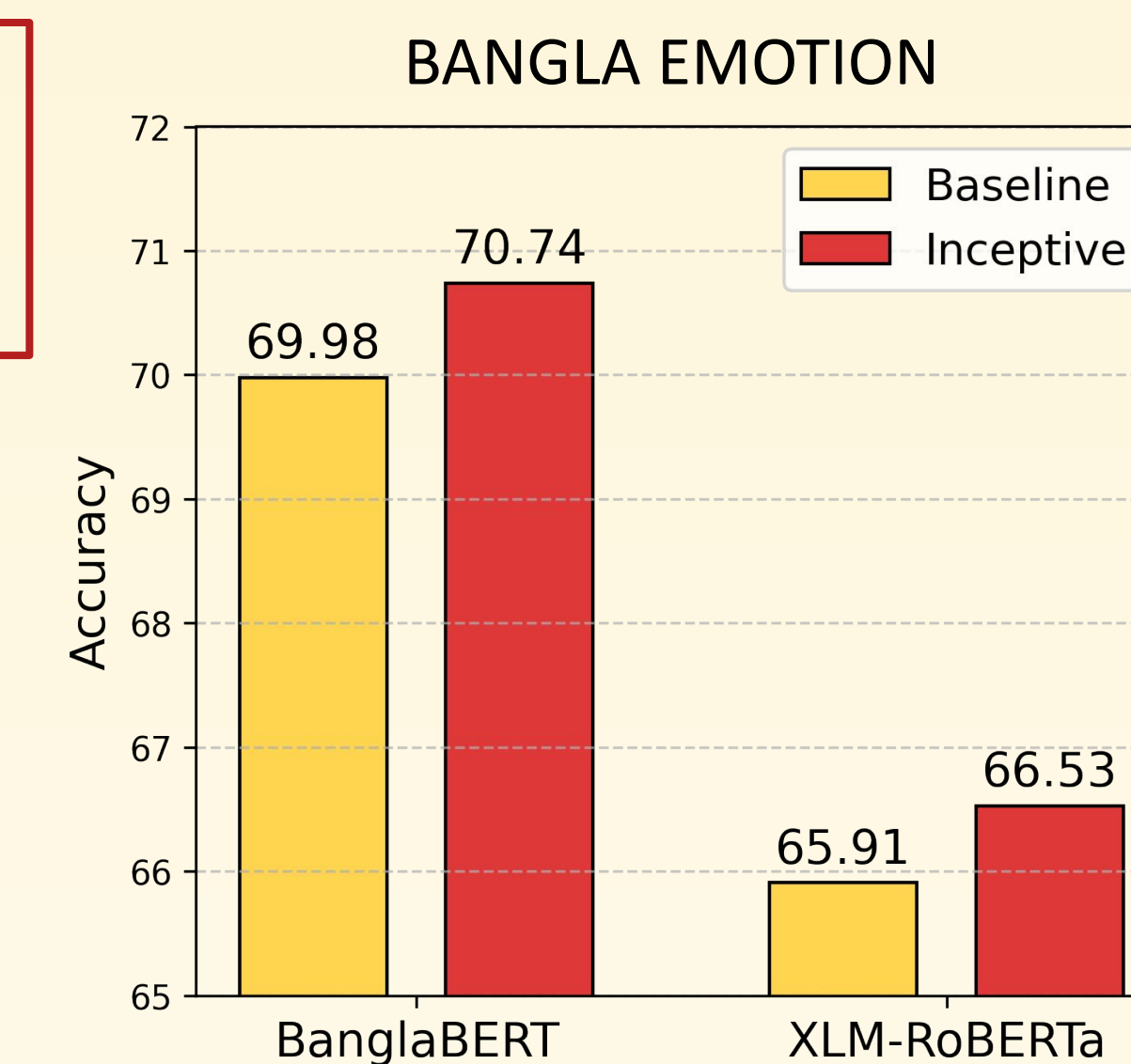
Statistically significant (Wilcoxon $p < 0.05$) Across 10 runs

10-fold CV results: Mean: incept > baseline Std dev: incept < baseline

Positive performance - inference trade-off in OHSUMED, CAVES, irony, comparable in emotion



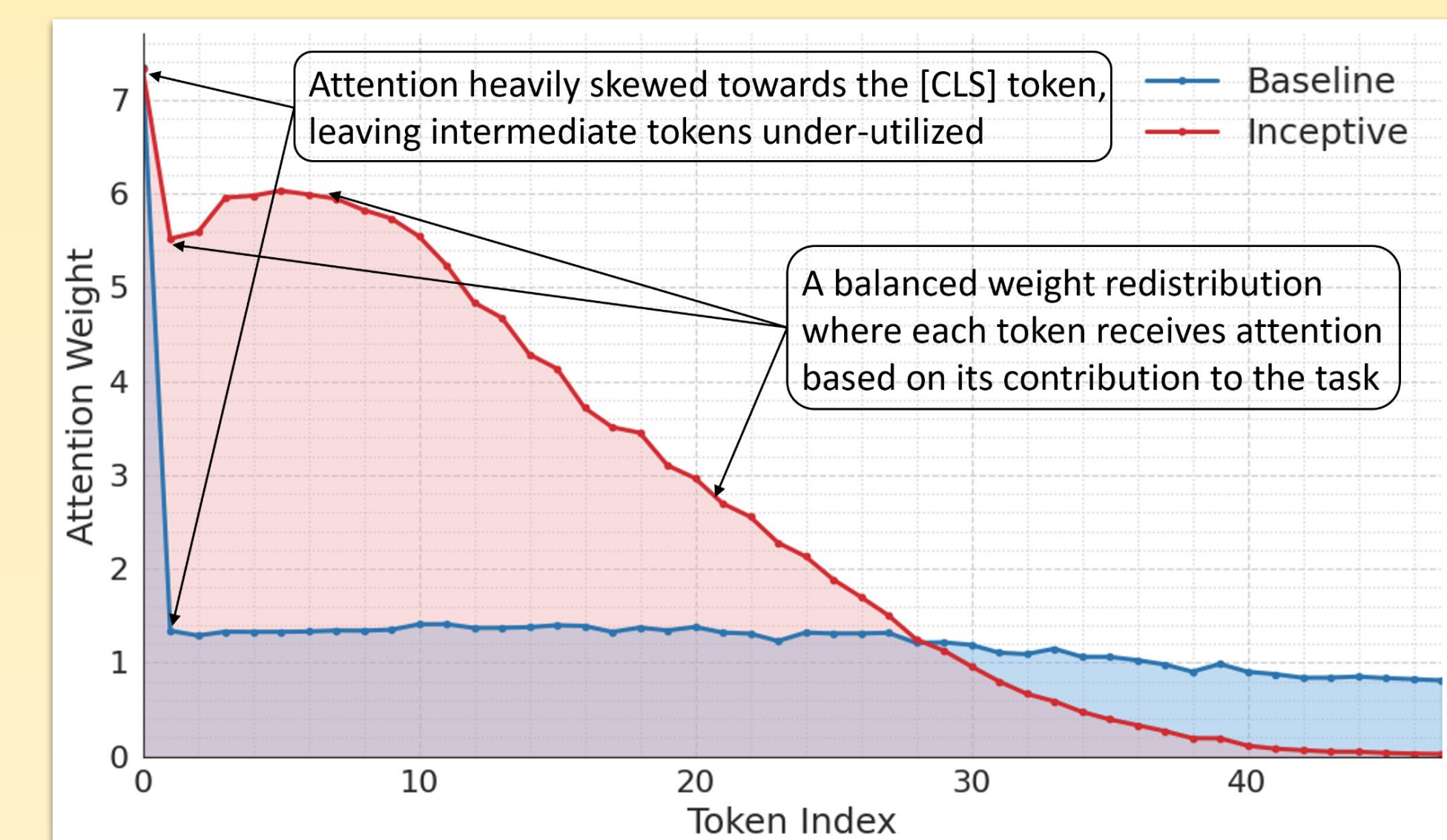
Ablation study using the best performing model. i stands for Inceptive.



Multi-scale convolution performs better than any single convolution branch

Post-convolution attention is necessary

Why it Helps



"work Wednesday-Sunday ... #yay #not #moneytho"

BERTweet fails to detect irony as the dominant neutral phrase outweighs the subtle sarcastic hashtags in the final [CLS] representation

Inceptive BERTweet correctly identifies irony by prioritizing the contradictory localized cues ('#yay', '#not') over the neutral phrase

Why it Matters

- Same architecture improves EIGHT different baselines
- Modular and plug-and-play by nature
- Maintains inference and memory efficiency
- Generalizable across models, domains and languages

Takeaways

- [CLS] token is a bottleneck in tasks that require nuanced understanding of local patterns (e.g., irony) or need to identify dispersed information in long texts (e.g., multi-label classification)
- Multi-scale convolution followed by self-attention increases token coverage and mitigates over-reliance on any single token
- There is no single 'best' kernel size - different kernel sizes capture complementary linguistic patterns, and their combination is what enables the model to adapt effectively across diverse domains and tasks